

IBM DATA ANALYST PROFESSIONAL CERTIFICATE

Capstone Project: Technology Trends Analysis

Extracting insights from the Stack Overflow Developer Survey

Priyanka Mishra • Data Analyst Portfolio
May 24, 2026

PRESENTATION OUTLINE



Strategic Context

Executive Summary and Introduction to the report's purpose and audience.



Methodology

Data sources, cleaning steps, and tools used for technical analysis.



Core Trends

In-depth analysis of Programming Languages and Databases (Current vs. Desired).



Demographics

Gender gap analysis and geographic distribution of the



Dashboard

Interactive visualization of current technology usage and



Final Outcome

Discussion of results, conclusions, and strategic

EXECUTIVE SUMMARY

- ✓ **Market Dominance:** JavaScript and HTML/CSS continue to lead current usage, but Python is rapidly closing the gap.
- ✓ **Future Trajectory:** TypeScript and Python show the highest "Desired" growth potential for next-year implementation.
- ✓ **Database Shift:** A clear move from legacy RDBMS like MySQL toward Open Source PostgreSQL and MongoDB.
- ✓ **Talent Landscape:** Significant gender disparity remains in IT, highlighting a critical need for diversity initiatives.
- ✓ **Salary Correlation:** High demand for Swift and Python correlates strongly with premium annual salary offerings.

 Professional Data Analyst Workspace

| INTRODUCTION & REPORT VALUE

 Abstract Network Background

Report Purpose

To identify emerging technology trends using the 2019 Stack Overflow Developer Survey, providing data-driven guidance for skill acquisition.

Target Audience

IT Professionals looking to pivot, HR Leaders identifying training needs, and Developers planning their learning roadmap.

Value Proposition

Strategic alignment of personal and corporate skills with global market demand to maximize ROI on education and hiring.

METHODOLOGY & TECHNICAL PROCESS



Data Sources

Integration of Stack Overflow 2019 Survey results, GitHub Job Posting APIs, and industry Salary datasets.



Data Wrangling

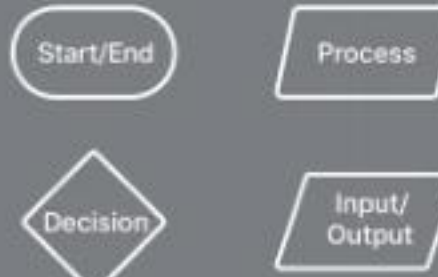
Cleaned 11,000+ entries using Python. Handled missing values through imputation and normalized salary data to a single scale.



Visualization

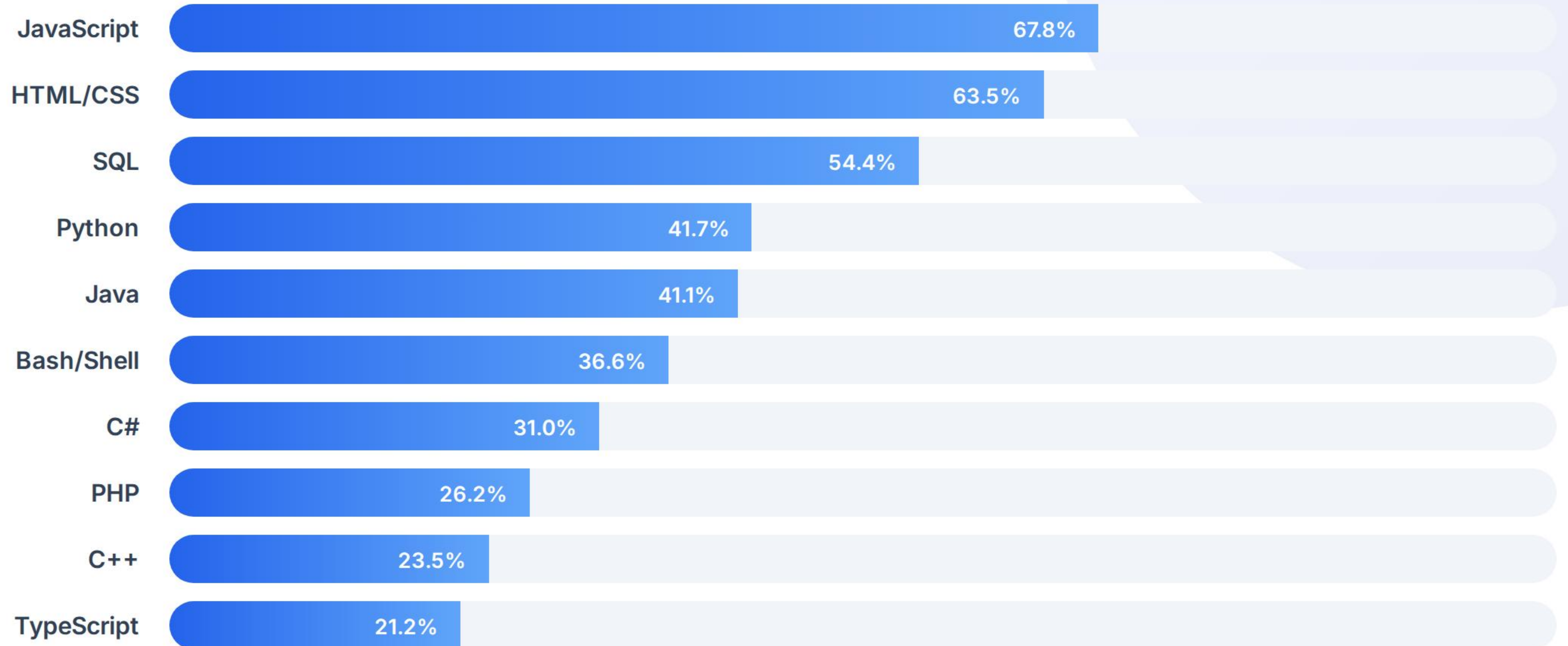
Utilized IBM Cognos Analytics for dashboarding and Matplotlib/Seaborn for exploratory trend analysis.

Flowchart Symbols



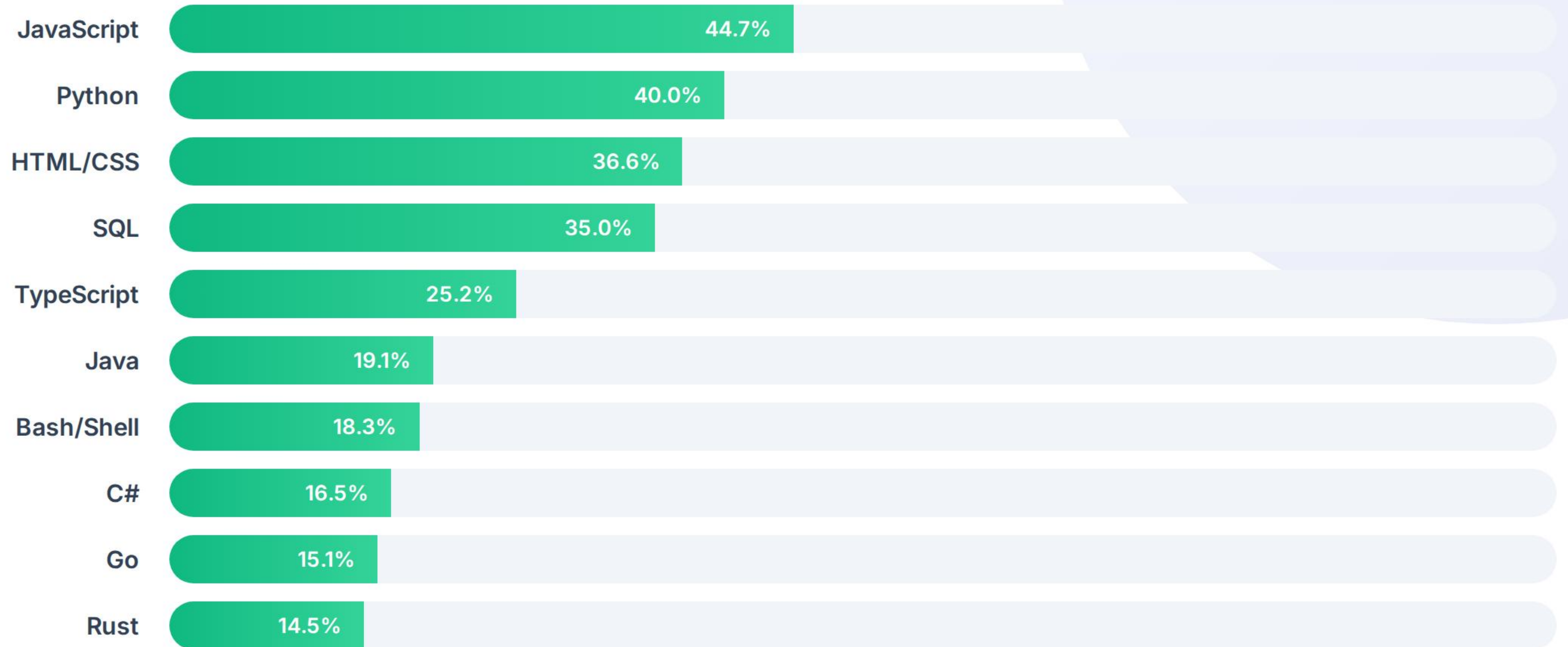
Flowchart AI

TOP 10 PROGRAMMING LANGUAGES: CURRENT USAGE



Survey data reflects respondents' current languages worked with in 2019.

TOP 10 LANGUAGES DESIRED FOR NEXT YEAR



PROGRAMMING TRENDS: FINDINGS

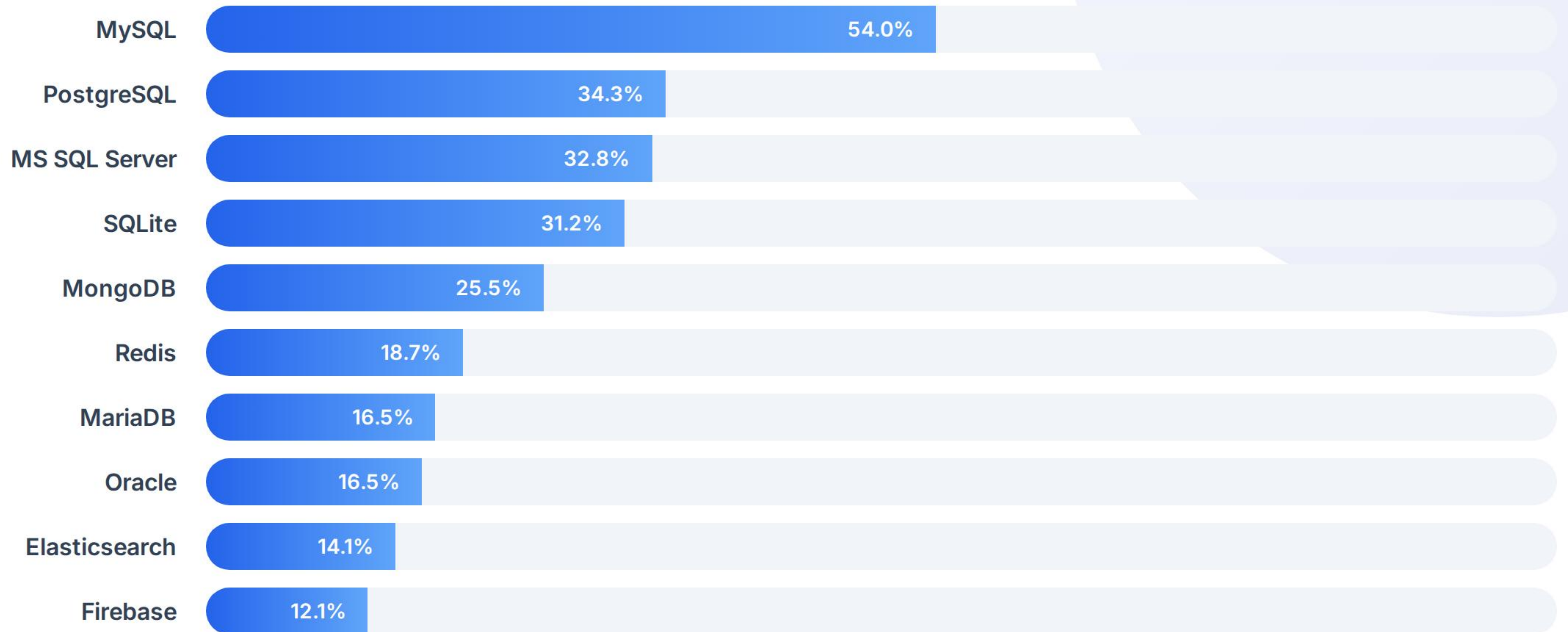
Key Findings

- ✓ JavaScript holds the top spot for both current and future years.
- ✓ Python is the fastest-growing major language in terms of desire.
- ✓ TypeScript enters the Top 5 desired languages, jumping from lower rankings.

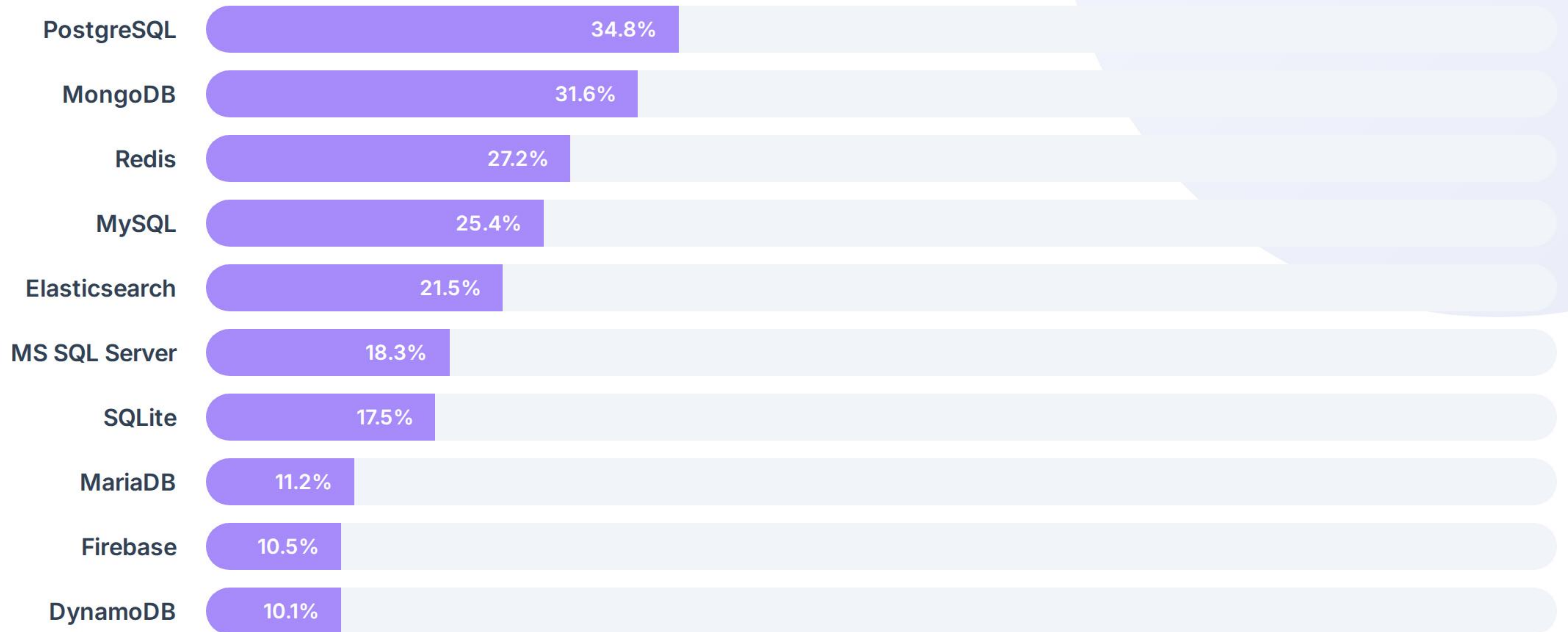
Implications

- ✓ **Versatility Wins:** Full-stack languages (JS, Python) are non-negotiable for modern developers.
- ✓ **Shift to Safety:** Rising interest in TypeScript and Rust indicates a move toward system reliability.

TOP 10 DATABASES CURRENTLY IN USE



TOP 10 DATABASES DESIRED FOR NEXT YEAR



DATABASE TRENDS: FINDINGS

Market Shift

- ✓ **PostgreSQL** is overtaking MySQL as the preferred relational choice for next year.
- ✓ NoSQL leaders like **MongoDB** and **Redis** show high momentum in developer interest.

Strategic Implications

- ✓ **Open Source Preference:** Proprietary systems (Oracle, MS SQL) are losing mindshare.
- ✓ **High-Scale Architecture:** Demand for Redis/Elasticsearch suggests focus on real-time data performance.

DEMOGRAPHICS & DIVERSITY INSIGHTS

92%

Male Dominance in IT

Addressing the Gender Gap

The survey reveals a critical imbalance: while 11% of US respondents are women (up from 9%), the global average remains starkly skewed. Diversity initiatives are no longer just social goals but economic imperatives for a healthy talent pipeline.

 Global Diversity Visual

DISCUSSION: THE FUTURE OF IT

"The speed of emerging technology requires a shift from static knowledge to a culture of continuous retraining."

— Key takeaway from trend analysis

Agility

Developers must pivot every 18-24 months as languages like Python and TypeScript evolve.

Openness

The transition toward open-source databases reflects a demand for community-driven innovation.

Equity

Eliminating the gender gap is essential to solving the global developer shortage.

FINAL CONCLUSIONS

- ✓ **Skill Dominance:** Proficiency in JavaScript and Python remains the safest career investment for 2026.
- ✓ **Database Evolution:** Master PostgreSQL and NoSQL (MongoDB) to align with enterprise infrastructure shifts.
- ✓ **Strategic Adaptability:** Organizations must foster inclusive environments to attract the 8% currently underserved female talent pool.
- ✓ **Technical Summary:** The project successfully integrated survey, API, and salary data to provide a holistic view of the tech landscape.



Questions & Discussion

Priyanka Mishra • Capstone Final Submission

IBM Data Analyst Professional Certificate

IMAGE SOURCES



Thumbnail

https://satechi.com/cdn/shop/articles/cubedock_lifestyle_2.webp?v=1778195924&width=3000

for

Source: satechi.com

satechi.com



https://static.vecteezy.com/system/resources/previews/025/872/117/non_2x/illustration-molecule-and-internet-connect-technology-on-dark-blue-background-abstract-internet-network-connection-design-for-web-site-digital-data-communication-science-futuristic-concept-vector.jpg

Source: www.vecteezy.com



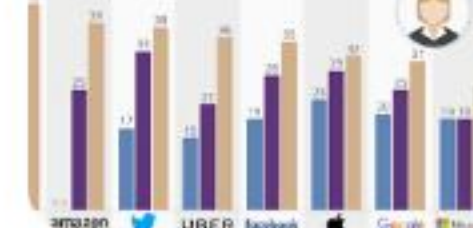
Thumbnail

<https://cdn.flowchartai.org/blog/images/flowchart-symbols.webp>

for

Source: flowchartai.org

flowchartai.org



<https://www.storybench.org/wp-content/uploads/2018/12/statista.png>

Source: www.storybench.org